How a macrophage *destroys* what it engulfs

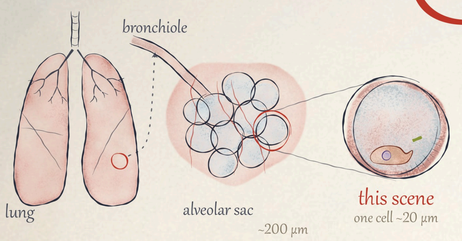
Cell Biology seminar · First-year Medicine

1 Aitor · 2 Nerea · 2.7 · 3 Urtzi · 4 Eneritz · 5 Nora · 6 Aitor

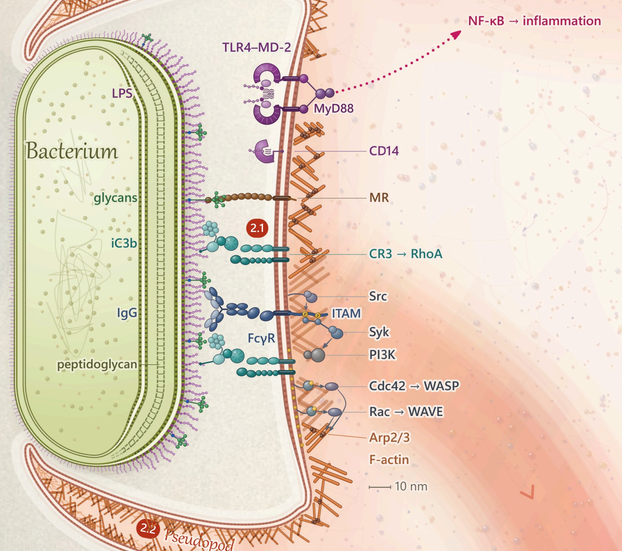
Phagocytic route · Enzyme route · Recycling
Membrane fusion · Pathological effect · Inflammatory signal
dashed = possible

Not to scale · molecular detail enlarged
cell ~20 µm · bacterium ~2 µm · receptor ~10 nm

1 Alveolar macrophage

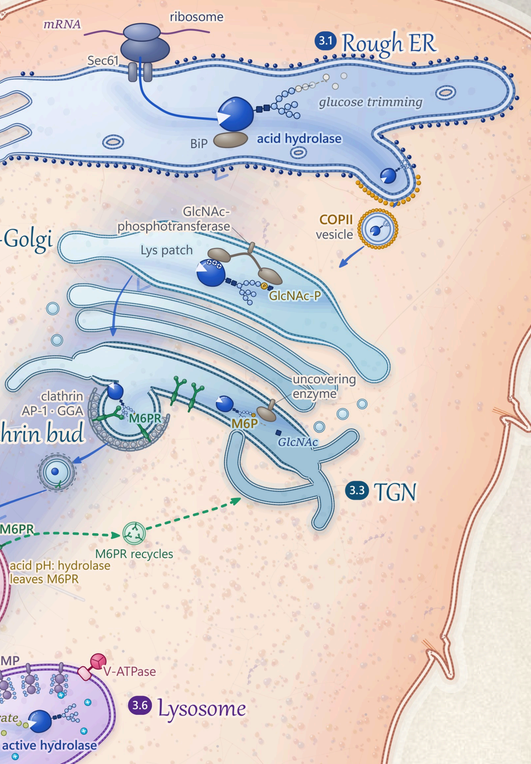


2 Recognition



3 Lysosome biogenesis

GlcNAc · mannose · glucose · phosphate



Nucleus

Rough ER

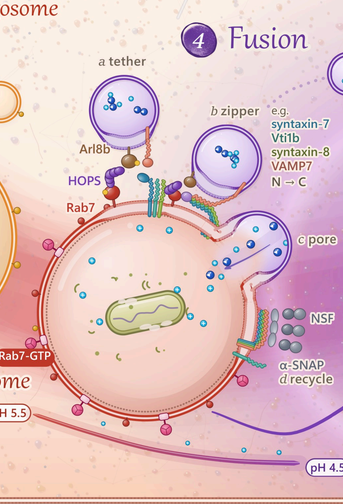
5 Particles

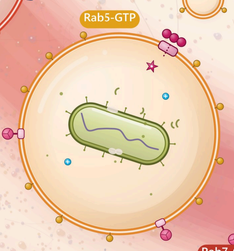
cleared

5.1 uptake
via MARCO / SR-A

2.3 Nascent phagosome

4 Fusion





2.4 Early phagosome

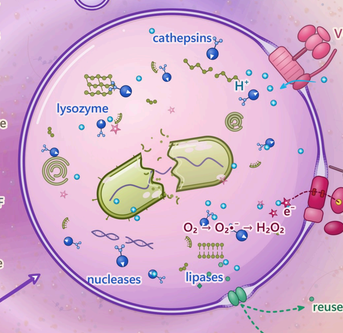
pH 6.5

2.5 Late phagosome

pH 5.5

pH 4.5

2.6 Phagolysosome



2.7 Killing

NOX2
gp91phox · p22phox
p47 · p67 · p40phox
Rac-GTP
NADPH → NADP+
burst starts at 2.3
iNOS: NO → ONOO-

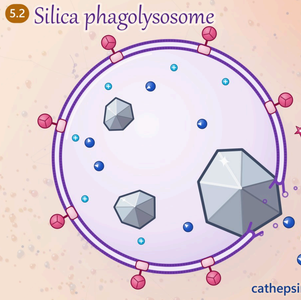
reuse

recycled

Mitochondrion

5.2 Silica phagolysosome

silicosis: crystalline-silica lung disease



possible re-uptake

signal 1 · priming (e.g. LPS)
NF-κB → NLRP3 + pro-IL-1β made

TLR
ROS
NF-κB
signal 2
K+ efflux
cathepsin B
NLRP3
pro-IL-1β / -18
possible lytic death
NLRP3 + ASC
inflammasome
caspase-1
IL-1β · IL-18
GSDMD pore
~20 nm
GSDMD → GSDMD-N
IL-1β · TNF-α

5.3 Asbestos

asbestosis: asbestos fibrosis

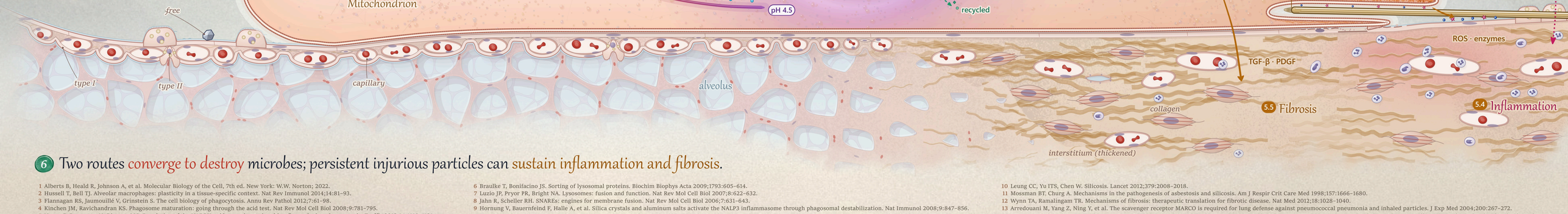


ROS · enzymes

TGF-β · PDGF

5.5 Fibrosis

5.4 Inflammation



6 Two routes converge to destroy microbes; persistent injurious particles can sustain inflammation and fibrosis.

1 Alberts B, Heald R, Johnson A, et al. Molecular Biology of the Cell, 7th ed. New York: W.W. Norton; 2022.
2 Hussell T, Bell TJ. Alveolar macrophages: plasticity in a tissue-specific context. Nat Rev Immunol 2014;14:81–93.
3 Flannagan RS, Jaumouillé V, Grinstein S. The cell biology of phagocytosis. Annu Rev Pathol 2012;7:61–98.
4 Kinchen JM, Ravichandran KS. Phagosome maturation: going through the acid test. Nat Rev Mol Cell Biol 2008;9:781–795.
5 Nunes P, Demaurex N, Dinauer MC. Regulation of the NADPH oxidase and associated ion fluxes during phagocytosis. Traffic 2013;14:1118–1131.
6 Braulke T, Bonifacino JS. Sorting of lysosomal proteins. Biochim Biophys Acta 2009;1793:605–614.
7 Lazzio JP, Pryor PR, Bright NA. Lysosomes: fusion and function. Nat Rev Mol Cell Biol 2007;8:622–632.
8 Jahn R, Scheller RH. SNAREs: engines for membrane fusion. Nat Rev Mol Cell Biol 2006;7:631–643.
9 Hornung V, Bauernfeind F, Halle A, et al. Silica crystals and aluminum salts activate the NALP3 inflammasome through phagosomal destabilization. Nat Immunol 2008;9:847–856.
10 Leung CC, Yu ITS, Chen W. Silicosis. Lancet 2012;379:2008–2018.
11 Mossman BT, Churg A. Mechanisms in the pathogenesis of asbestosis and silicosis. Am J Respir Crit Care Med 1998;157:1666–1680.
12 Wynn TA, Ramalingam TR. Mechanisms of fibrosis: therapeutic translation for fibrotic disease. Nat Med 2012;18:1028–1040.
13 Arredouani M, Yang Z, Ning Y, et al. The scavenger receptor MARCO is required for lung defense against pneumococcal pneumonia and inhaled particles. J Exp Med 2004;200:267–272.

Successive stages in one schematic cell · not to scale; molecular detail enlarged